Seth Balkrishan Memorial

BHILAI INSTITUTE OF TECHNOLOGY, DURG

(An Autonomous Institute affiliated to CSVTU, Bhilai)

SCHEME OF TEACHING AND EXAMINATION (Effective from 2020-2021 Batch)

B.Tech. (Information Technology) Sixth Semester

Sl. No.	Board of Studies (BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
							Theory/Lab				
				L	T	P	ESE	CT	TA		
1.	Information Technology	Principles of Communication System	106601IT	3	1	-	100	20	30	150	4
2.	Information Technology	Artificial Intelligence	106602IT	2	1	-	100	20	30	150	3
3.	Humanities	Employment Communication	100603HM	3	-	-	100	20	30	150	3
4..	Information Technology	Professional Elective-1	Refer Table-1	3	-	-	100	20	30	150	3
5.	Refer Table-2	Open Elective-1	Refer Table-2	3	-	-	100	20	30	150	3
6.	Information Technology	Principles of Communication System Lab	106691IT	-	-	2	25	-	25	50	1
7.	Information Technology	Artificial Intelligence Lab	106692IT	-	-	2	25	-	25	50	1
8.	Information Technology	Android Lab	106693IT	-	-	2	25	-	25	50	1
9.	Information Technology	Minor Project – II	106694IT	-	-	2	50	-	25	75	1
10.	Humanities	Essence of Indian Knowledge Tradition	100695HM	-	-	-	-	-	25	25	-
Total				**14**	**2**	**8**	**625**	**100**	**275**	**1000**	**20**

Table-1 : Program Elective – 1

S. No.	Board of Study	Course (Subject)	Course Code
1.	Information Technology	Computer Graphics & Data Visualization	106621IT
2.	Information Technology	Microcontroller & Embedded Systems	106622IT
3.	Information Technology	Data Mining	106623IT
4.	Information Technology	Digital Image Processing	106624IT
5.	Information Technology	Natural Language Processing	106625IT

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Table 2 : Open Elective -I					
S. No.	**Board of Study**	**Courses (Subject)**	**Course Code**	**Eligible Branches**	**Course Link for Reference**
1.	Civil Engineering	Solid and Hazardous Waste Management	100641CE	Open to all branches	https://archive.nptel.ac.in/courses/105/103/105103205/
2.	Computer Science & Engineering	Data Structures and Algorithms	100642CS	Open to all branches except CSE & IT	https://archive.nptel.ac.in/courses/106/102/106102064/
3.	Computer Science & Engineering	Unix Operating System	100643CS	Open to all branches except CSE & IT	https://archive.nptel.ac.in/courses/117/106/117106113/
4.	Electrical & Electronics Engineering	Renewable Energy Engineering	100644TE	Open to all branches except EE	https://archive.nptel.ac.in/courses/103/103/103103206/
5.	Electrical Engineering	Electric Vehicles	100645EE	Open to all branches	https://archive.nptel.ac.in/courses/108/106/108106170/
6.	Electronics &Telecommunication Engineering	IoT Applications	100646ET	Open to all branches except CSE & IT	https://archive.nptel.ac.in/courses/106/105/106105166/
7.	Information Technology	Programming in PHP and MySQL	100647IT	Open to all branches except CSE & IT	https://www.youtube.com/playlist?list=PL7WFbgpeASD3kmW7h5qhGZGIWTOXGGbTZ
8.	Mechanical Engineering	Operation Research	100648ME	Open to all branches	https://archive.nptel.ac.in/courses/110/106/110106062/
9.	Management	Financial Management	100649MG	Open to all branches	https://archive.nptel.ac.in/courses/110/107/110107144/
10.	Computer Applications	Python	100650CA	Open to all branches	https://archive.nptel.ac.in/courses/106/106/106106182/

Note:

(a) Abbreviations used : L-Lecture, T- Tutorial, P- Practical, ESE- End Semester Exam, CT- Class Test, TA- Teacher's Assessment

(b) 1/4th of total strength of students subject to minimum of 20 students is required to offer an elective in the department in a particular academic session.

(c) Choice of elective course once made for an examination cannot be changed in future examinations.

(d) The duration of end semester examination of all theory papers will be of three hours.

(e) "Employment Communication" and "Essence of Indian Knowledge Tradition" will be conducted by the Humanities / relevant discipline as decided by the Principal.

SYLLABUS

B.TECH. (INFORMATION TECHNOLOGY)

SIXTH SEMESTER

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SYLLABUS

B.Tech. (Information Technology) Sixth Semester

PRINCIPLES OF COMMUNICATION SYSTEM					
Course Code	**106601IT**	**L = 3**	**T = 1**	**P = 0**	**Credits = 4**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
Objective is to make students understand the principles of communication system, to setup the strong foundation for the subjects such as Information Theory and coding, cryptography, network security, cellular and wireless communication.	**On successful completion of the course, the student will be able to:** **CO1:** Analyze the concepts of amplitude modulation and demodulation for transmission and reception of Information signal. **CO2:** Apply the concepts of angle modulation for transmission, receptions of information signal and can explain the better noise performance of FM compared to AM. **CO3:** Breakdown the process of analog to digital conversion and compare different techniques as per different requirements in each process. **CO4:** Select and predict the requirement of different digital modulation techniques on the basis of bandwidth, power efficiency and high data rate for different applications. **CO5:** Identify and justify the requirements of satellite and optical communication in recent trends.

UNIT I: Amplitude Modulation **(CO1)**

Need for Modulation, Amplitude Modulation, Amplitude Modulation Index, Modulation Index for Sinusoidal AM, Frequency spectrum for Sinusoidal AM, Average power for Sinusoidal AM, Effective voltage and current for sinusoidal AM, Balanced Modulator, The Square law demodulator, DSBSC Modulation, SSB modulation and generation, VSB, FDM. Noise in communication systems, Signal to noise ratio Noise performance in AM. **[10 Hrs]**

UNIT II: Angle Modulation **(CO2)**

Phase and frequency modulation and their relationship. Frequency deviation, spectrum of FM Signal, BW of FM Signal, Effect of modulation on BW, constant BW, FM phasor diagram, Narrow-band F.M. Armstrong and Parameter variation methods of FM generation and FM demodulators. Noise performance in FM and comparison with AM. **[10 Hrs]**

UNIT III: Baseband Modulation **(CO3)**

Sampling theorem, Pulse Modulation: PAM, PPM, PWM. Quantization of Signals, Quantization error, TDM, Pulse Code Modulation (PCM), DPCM, DM, ADM and their comparative performance evaluation. **[10 Hrs]**

UNIT IV: Digital Modulation Techniques **(CO4)**

Modulation techniques for ASK, QASK, FSK, M-array FSK, BPSK, DPSK, QPSK, M-array PSK, QAM. Comparison of Noise performance of various PSK and FSK systems. Theme Example Orthogonal Frequency Division multiplexing (OFDM). **[10Hrs]**

UNIT V: Advanced Communication Techniques **(CO5)**

Satellite Communication: Satellite orbits and positioning, Satellite Communication Systems, Satellite Subsystems, Ground Stations, Satellite Applications, Global Navigation Systems.

Fiber Optic Communication: Optical Principles, Optical Communication Systems, Fiber Optic Cables, Optical Transmitter and Receivers, Wavelength Division Multiplexing, Passive Optical Networks, 40/100 Gbps Networks and Beyond. **[8Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Principles of Communication system	H.Taub and D.L. Shiling	TMH
2.	Communication Systems	R. P. Singh and S. D. Sapre	TMH
3.	Principles of Electronic Communication Systems	Louis E. Frenzel Jr.	McGraw Hill

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Electronic Communications	Roddy & Coolen	PHI
2.	An Introduction to the Principle of Communication	J.C. Hancock	McGraw Hill
3.	Communication System	A.B. Carlson	McGraw Hill
4.	Electronic Communication System	Kenedy & Davis	TMH

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B.Tech. (Information Technology) Sixth Semester

ARTIFICIAL INTELLIGENCE					
Course Code	**106602IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to make the student understand the concept of principles of AI towards problem solving, inference, perception, knowledge representation and learning and also to have proficiency in a traditional AI language including an ability to write simple to intermediate programs and an ability to understand code written in that language.	**On successful completion of the course, the student will be able to:** **CO1**: Apply heuristics functions for artificial intelligence (AI) and expert systems. **CO2:** Implement knowledge representation techniques on real-time problems and inference knowledge from given information. **CO3**: Identify uncertainty and indicate appropriate AI techniques for intelligent agents, expert systems, artificial neural networks and other machine learning models. **CO4**: Use the proficiency in applying scientific methods to models of Pattern Recognition. **CO5:** Demonstrate proficiency in making Expert Systems according to the needs of society.

UNIT I:Introduction **(CO1)**

Introduction to AI, Problem Solving, State space search, Blind search: Depth first search, Breadth first search, Informed search: Heuristic function, Hill climbing search, Best first search, A* & AO* Search, Constraint satisfaction. **[7Hrs]**

UNIT II: Knowledge Representation (KR) **(CO2)**

Introduction to KR, Knowledge agent, Predicate logic, WFF, Inference rule & theorem proving forward chaining, backward chaining, resolution; Propositional knowledge, Boolean circuit agents. Rule Based Systems, Forward reasoning: Conflict resolution, backward reasoning: Use of Backtracking. **[8Hrs]**

UNIT III: Reasoning in uncertain environments **(CO3)**

Reasoning in uncertain environments, Fuzzy logic, fuzzy composition relation, operations on fuzzy sets. Probabilistic reasoning, Markov processes and Hidden Markov models. **[7Hrs]**

UNIT IV: Pattern Recognition **(CO4)**

Introduction, Design principles of pattern recognition system, Statistical Pattern recognition, Parameter estimation methods-Principle Component Analysis (PCA) and Linear Discriminant Analysis (LDA), Classification Techniques-Nearest Neighbor (NN) Rule, Support Vector Machine (SVM), K-means clustering. **[7Hrs]**

		August 2022	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2022-23 Onwards

UNIT V: Expert System & AI languages **CO5**

Need & Justification for expert systems- cognitive problems, Expert System Architectures, Rule based systems, Non production system, knowledge acquisition, Case studies of expert systems. AI language: Prolog syntax, Programming with PROLOG, LISP syntax, LISP programming. **[7Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Artificial Intelligence	Elaine Rich and Kevin Knight	Tata McGraw Hill
2.	Introduction to Artificial Intelligence and Expert Systems	Dan W. Patterson	Tata McGraw Hill

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Principles of Artificial Intelligence	Nils J. Nilsson	Narosa Publishing house
2.	Programming in PROLOG	Clocksin & C.S. Melish	Narosa Publishing house

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SYLLABUS

B.Tech. (Common to All Branches) Sixth Semester

EMPLOYMENT COMMUNICATION					
Course Code	**100603HM**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course intends to effectively train and prepare students to join workforce and function constructively. It aims to teach basic strategies of cracking GD's to demonstrate their team working abilities so that they can emerge as leaders in achieving organizational goals	**On successful completion of the course, the student will be able to:** **CO1:** Demonstrate a better understanding about the importance of communication especially professional communication and clearly follow the procedure and etiquettes involved in it. **CO2:** Employ positive messages both verbally and non-verbally in order to function constructively at work place. **CO3:** Draft impressive CV and Resume aimed to grasp the attention of employers. **CO4:** Exhibit team work and other employability skills by performing effectively in Group Discussion and Personal Interview. **CO5:** Demonstrate better life skills which eventually support healthy work culture.

UNIT – I Introduction **CO1**

- Employment Communication: *Introduction*
- Introduction to Communication: *What is communication; Importance of Communication, Functions of Communication*
- Communication Skills at work place: *Basic purpose of communication at workplace*, *Employability skills; Planning and Inventing Strategies; Strategies for teamwork and workplace communication*
- Inter-cultural Communication: *Preparing to communicate across cultures, Demographic diversity, Reducing strangeness of strangers; Technology and inter-cultural communication.* **[8Hrs]**

UNIT – II Verbal & Non-Verbal Communication **CO2**

- Verbal Communication: *Language and meaning; How language works; Patterns of miscommunication; Developing Language Skills*
- Non-Verbal Communication: *Communicating without words; characteristics and functions of non-verbal communication, cultural diversities*
- Body Language: *Dress up, Posture, Gesture, Facial Expression, Eye contact, Voice Modulation*
- Listening Skills: *Types of Listening*; *Components and Essentials of Active Listening; Decoding and its deterrents; Role of Critical Thinking; Feedback.* **[8Hrs]**

UNIT – III Employment Process I **CO3**

- Introduction to Employment Process: *Searching for employment and writing employment messages; Strategies in job search process. Identifying appropriate job*
- CV writing: *Writing Cover letters, Types of cover letters, Cultural considerations*
- Resume Writing: *Paper Resume and Scannable Resume, Resume Components, Common Resume Problems.* **[7Hrs]**

UNIT – IV Employment Process II **CO4**

- Group Discussion: *Definition, Qualities looked for in GD*, *Types of GD; Strategies and Techniques to initiate and perform effectively; Body Language; Do's and Don'ts*
- Interviewing for Employment: *Preparing for Personal Interview; Assessing Interview anxiety; Guidelines to improve effectiveness in Interviews*
- Taking Questions during Personal Interview: *Types of Questions; Effective Listening; Preparedness to answer strategically; Responsibilities in an Interview; Behavioral Planning and Pattern; Handling negative or illegal questions; Video and Telephonic Interviews.* **[7Hrs]**

UNIT – V Life Skill **CO5**

- Life/Social Skill: *Meditation, Spirituality, How to meditate.* **[6Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Strategies for Engineering Communication	Susan Stevenson & Steve Whitmore	Willey India Pvt Ltd.
2.	Group Discussion: Theory & Technique	Harnack, Fest and Jones	Prentice Hall
3.	The Art of Winning at Interviews	Nagasudha Ravinuthala	ICFAI University Press

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Secrets of Power Presentations	Peter Urs Bender	The Achievement Group Toronto
2.	The Skills of Interviewing: A guide for Managers & Trainers	Leslie Rae	Nichols Pub Co
3.	Presenting with Power	Captivate, Motivate, Inspire & Persuade	How to Books
4.	Course on Employment Communication	https://archive.nptel.ac.in/courses/109/105/109105144	

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B.Tech. (Information Technology) Sixth Semester

COMPUTER GRAPHICS & DATA VISUALIZATION					
Course Code	**106621IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to make the student understand the fundamental concepts of graphics and data visualization, and apply the acquired knowledge pertaining to 2D and 3D concepts in graphics programming.	**On successful completion of the course, the student will be able to:** **CO1:** Enumerate the functionalities of coordinate systems pertaining to graphics manipulation and implementation of different algorithms related to graphical objects. **CO2:** Design and demonstrate the 2D and 3D objects using graphics algorithms. **CO3:** Model and render objects by comprehending the complexities of illumination in virtual scenes along with projection. **CO4:** Identify the different data types, visualization types to bring out the insight. Relate the visualization towards the problem based on the dataset. **CO5**: Compare the different attributes and showcasing them in plots. Identify and create various visualizations for geospatial and table data.

UNIT I: : Introduction **(CO1)**

Basic Concepts & Techniques: Pixel, Coordinate Representations, Raster and Random generator. Graphics Functions, RGB Color Component, Bresenham's Line drawing Algorithm. Midpoint Circle Algorithm, Bresenham's Circle Algorithm. Midpoint Ellipse Algorithm. Pixel Addressing and Object Geometry. Solid area filling Algorithm: Boundary, Flood fill algorithm. **[7 Hrs]**

UNIT II: Viewing and Clipping **(CO2)**

Windows and Viewport, Viewing transformation; Clipping of 2-D & 3-D Lines: Cyrus Beck, Cohen Sutherland clipping, Polygon clipping: Hodgeman-Sutherland polygon clipping. **[8 Hrs]**

UNIT III: Curves. Projection and Surfaces **(CO3)**

Curve representation; Parametric Spline curve; Bezier Curve: parametric continuity (c0, c1, c2), Geometric continuity, Bernstein polynomials, Condition for smooth joining of 2 curve segments, Convex Hull property, B-Spline Curves: Knot vectors (uniform and open uniform curves, non-uniform curve), basis function. Parallel Projection: Orthographic, Axonometric and Oblique Projection; Perspective Projection: Vanishing point, 1,2& 3 perspective projection. Visible-Surface Detection Methods: Back-Face Detection, Depth-Buffer Method. Ray-Casting Method. Basic Illumination Models: Ambient Light, Diffuse Reflection, Specular Reflection and the Phong Model. **[7 Hrs]**

UNIT IV: Introduction to Data Visualization **(CO4)**
Overview of data visualization: - Data Abstraction -Analysis: Four Levels for Validation- Task Abstraction - Analysis: Four Levels for Validation, Visualization Techniques. **[7 Hrs]**

UNIT V: Visual Analytic **(CO5)**
Visual Variables- Networks and Trees - Map Color and Other Channels- Manipulate View, Visualization Tools and Techniques: Introduction to data visualization tools. **[7Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Computer Graphics with OPENGL - C Version	Donald Hearn, Pauline Baker	Pearson Education
2.	Visualization Analysis and Design	Tamara Munzer	CRC Press

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Computer Graphics: Principles and Practice	John F. Hughes, Andries Van Dam, Morgan Mc Guire ,David F. Sklar , James D. Foley, Steven K. Feiner and Kurt Akeley	Addison Wesley Professional
2.	Data Visualization Principles and Practice	Alexandru Telea	CRC Press

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B.Tech. (Information Technology) SixthSemester

MICROCONTROLLER & EMBEDDED SYSTEMS					
Course Code	**106622IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The objective of this course is to impart knowledge about the microcontrollers, its programming, interrupts timers and assembly language. The concepts of ARM architecture and real-time operating system. To provide experience to integrate hardware and software for microcontroller application system. To impart ability to put together processor, peripherals and memory and build a real time system.	**On successful completion of the course, the student will be able to:** **CO1:** Interpret interfacing and core expertise knowledge in microcontroller **CO2:** Write the programs for microcontrollers **CO3:** Explain the concepts of embedded systems **CO4:** Discuss the design concept of embedded systems and the role of embedded systems in industry **CO5:** Describe the architecture and programming of ARM processor

UNIT 1 **(CO1)**

Microcontroller: Introduction to Microcontrollers, Evolution, Architectures, Implementations, Background and History of Embedded Systems, Characteristics of ES, Hardware/Software Co-Design, RISC vs CISC, MCS-51 Family Overview, Important Features, Architecture, 8051 Pin Functions, Architecture, Addressing Modes, Instruction Set, Instruction Types, Applications of ASIC and FPGA in ES. **[8Hrs]**

UNIT II **(CO2)**

Programming: Assembly Programming, Timer Registers, Timer Modes, Overflow Flags, Clocking Sources, Timer Counter Interrupts, Baud Rate Generation, Serial Port Register, Modes of Operation, Processing Interrupts, Interrupt Service Routines, Look-up Tables. **[7Hrs]**

UNIT III **(CO3)**

Embedded Software Development: Software development flow, Polling, Interrupt driven, Multitasking systems, Architecture of an RTOS, Important features of RTOS, Embedded Systems Programming, Locks and Semaphores, Operating System Timers and Interrupts, Exceptions, Tasks, Task states and scheduling, Task structures, Synchronization, Communication and concurrency, Semaphores, Real-time clock. **[7Hrs]**

UNIT IV **(CO4)**

32-Bit Cortex-M Architecture: CPU architecture, Memory model, Registers, Modes, Exceptions, Interrupts, Exception handlers, interrupt controllers, Power modes, Hardware features and optimizations, Advanced bus standards like AMBA, The NVIC on ARM Cortex-M. **[7Hrs]**

		August 2022	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2022-23 Onwards

UNIT V **(CO5)**

Instruction Set of ARM: Syntax, Addressing modes and operands, Memory access instructions, Logical operations, Shift operations, Arithmetic operations, Stack, Functions and control flow, Assembler directives, Thumb and arm instruction differences, Development with Keil and Mbed, Applications like IoT and machine learning with cortex-M. **[7Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	The 8051 Microcontroller and Embedded Systems	Mazidi Muhammad Ali	Pearson Publication
2.	The Definitive Guide to ARM Cortex-M3 processors	Joseph Yiu	Newnes publication
3	Introduction to ARM Cortex-M Microcontrollers	Jonathan W. Valvano	Create Space

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Embedded System	R. Kamal	McGraw Hill Education
2.	Real-Time Interfacing to ARM Cortex-M Microcontrollers	Jonathan W. Valvano	Create Space
3.	Computer Organization and Design: The Hardware/Software interface	D. A. Patterson and J. H. Hennessy	Morgan Kaufman Publishers
4.	Real-Time Operating Systems for ARM Cortex-M Microcontrollers	Jonathan W. Valvano	Create Space

Useful Links.

1. https://nptel.ac.in/courses/108/107/108107029/ Dr. PramodAgarwal, IIT Roorkee.
2. https://nptel.ac.in/courses/106/108/106108100/ Prof. Krishna Kumar, IISc Bangalore.
3. https://nptel.ac.in/courses/117/104/117104072/ Dr. S.P. Das, IIT Kanpur

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B.Tech. (Information Technology) Sixth Semester

DATA MINING					
Course Code	**106623IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
• To provide students with basic concepts and techniques of Data Mining and Data Preprocessing and to provide a comprehensive introduction to Data Warehousing and knowledge discovery.	**After successful completion of this course, students will be able to:** **CO1:** Identify the type of data and apply preprocessing methods and data mining techniques accordingly. **CO2:** Design Data warehouse schemas for any organization. **CO3:** Generate interesting patterns from large amounts of data using association rule mining algorithms **CO4:** Evaluate the accuracy of supervised and unsupervised models and algorithms. **CO5:** Explore recent trends in data mining such as web mining, spatial-temporal mining.

UNIT I:Introduction to Data Mining and Data Preprocessing (CO1)

Introduction to Data mining: Motivation for Data Mining, its importance, Data Mining functionalities: Concept/Class Description: Characterization and Discrimination, Mining Frequent Patterns, Associations, and Correlations, Classification and Prediction, Cluster Analysis, Outlier Analysis, Evolution Analysis, Major issues in Data Mining. Major Tasks in Data Preprocessing: Data Cleaning, Data Transformation, Data Integration, Data Reduction and Data Discretization. **[7 Hrs]**

UNIT II: Data Warehousing, On-Line Analytical Processing and Knowledge Discovery (CO2)

Need for Data Warehousing, basic elements of data warehousing, Characteristics of Data Warehouse, Data Warehousing and OLAP technology, Principles of dimensional modeling, Dimensional modeling, Data Cubes, Schemas for Multidimensional Databases: Stars, Snowflakes, and Fact Constellations. Data Warehouse Architecture: Steps for the Design and Construction of Data Warehouses, A Three-Tier Data Warehouse Architecture, Data Warehouse Back-End Tools and Utilities, Metadata Repository, Types of OLAP Servers: ROLAP versus MOLAP versus HOLAP. Knowledge Discovery: KDD Process. **[7 Hrs]**

UNIT III: Mining Frequent Patterns, Associations, and Correlations (CO3)

Market Basket Analysis: Frequent Itemsets, Closed Itemsets and Association Rules, Association Rule Mining: The Apriori Algorithm: Finding Frequent Itemsets Using Candidate Generation, Generating Association Rules from Frequent Itemsets, Improving the Efficiency of Apriori. Mining Frequent Itemsets without Candidate Generation. Advanced Association Rule Techniques, Measuring the quality of rules. **[8 Hrs]**

		August 2022	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2022-23 Onwards

UNIT IV Supervised Techniques: Classification and Prediction **(CO4)**

Issues Regarding Classification and Prediction: Preparing the Data for Classification and Prediction, Classification by Decision. Tree Induction: Decision Tree Induction, Attribute Selection Measures, Tree Pruning, Scalability and Decision Tree Induction, Bayesian Classification: Bayes' Theorem, Naïve Bayesian Classification, Rule-Based Classification: Using IF-THEN Rules for Classification.

Unsupervised Techniques: Clustering, Cluster Analysis, Partitioning Methods, Hierarchical Methods, Density-based Methods, Grid based Methods, Outlier Detection. **[7Hrs]**

UNIT V: Advanced Techniques **(CO5)**

Mining Complex Data, Web Mining: Web Content Mining, Web Structure Mining, Web Usage Mining. Text Mining, Spatial Mining, Temporal mining, Applications and Trends in Data Mining. **[7 Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Data Mining: Concepts & Techniques ǁ	Jiawei Han & Micheline Kamber	Morgan Kaufman Publisher

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Data warehousing- concepts, Techniques, Products and Applications	C.S.R. Prabhu	Prentice Hall of India
2.	Data Mining Introductory and Advanced Topics	Margaret H. Dunham	Pearson Education
3.	Data Warehousing, Data Mining, & OLAP	Alex Berson, Stephen J. Smith	Tata McGraw Hill
4.	Insight into Data Mining: Theory & Practice	Soman K. P., Diwakar Shyam, Ajay, V.	Prentice Hall of India

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SYLLABUS

B.Tech. (Information Technology) Sixth Semester

DIGITAL IMAGE PROCESSING					
Course Code	**106624IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To make students learn about digital image fundamentals, simple image enhancement techniques in Spatial and Frequency domain, concepts of degradation function and restoration techniques, image segmentation and representation techniques and image compression and recognition methods.	**On successful completion of the course, the student will be able to:** **CO1:** Apply fundamentals of digital image processing, such as digitization, sampling, quantization, and 2D-transforms. **CO2:** Operate on images using the techniques of smoothing, sharpening and enhancement. **CO3:** Discuss the restoration concepts and filtering techniques. **CO4:** Apply the basics of segmentation, features extraction on images. **CO5:** Analyze and apply compression and recognition methods for color models.

UNIT I: DIGITAL IMAGE FUNDAMENTALS **(CO1)**

Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels - Color image fundamentals - RGB, HSI models, Two-dimensional mathematical preliminaries, 2D transforms - DFT, DCT. **[8Hrs]**

UNIT II: IMAGE ENHANCEMENT **(CO2)**

Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering– Smoothing and Sharpening Spatial Filtering, Frequency Domain: Introduction to Fourier Transform– Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters, Homomorphic filtering, Color image enhancement. **[7Hrs]**

UNIT III: IMAGE RESTORATION **(CO3)**

Image Restoration - degradation model, Properties, Noise models – Mean Filters – Order Statistics – Adaptive filters – Band reject Filters – Band pass Filters – Notch Filters – Optimum Notch Filtering – Inverse Filtering – Wiener filtering. **[7Hrs]**

UNIT IV IMAGE SEGMENTATION **(CO4)**

Edge detection, Edge linking via Hough transform – Thresholding - Region based segmentation – Region growing – Region splitting and merging – Morphological processing- erosion and dilation, Segmentation by morphological watersheds – basic concepts – Dam construction – Watershed segmentation algorithm.

Unsupervised Techniques: Clustering, Cluster Analysis, Partitioning Methods, Hierarchical Methods, Density-based Methods, Grid based Methods, Outlier Detection. **[7Hrs]**

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UNIT V: IMAGE COMPRESSION AND RECOGNITION (CO5)

Need for data compression, Huffman, Run Length Encoding, Shift codes, Arithmetic coding, JPEG standard, MPEG. Boundary representation, Boundary description, Fourier Descriptor, Regional Descriptors – Topological feature, Texture - Patterns and Pattern classes - Recognition based on matching. **[7Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Digital Image Processing	Rafael C. Gonzalez, Richard E. Woods	Pearson
2.	Fundamentals of Digital Image Processing	Anil K. Jain	Pearson

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Digital Image Processing	Kenneth R. Castleman	Pearson
2.	Digital Image Processing using MATLAB	Rafael C. Gonzalez, Richard E. Woods, Steven Eddins	Pearson Education, Inc
3.	Multidimensional Digital Signal Processing	D,E. Dudgeon and R. M. Mersereau	Prentice Hall
4.	Digital Image Processing	William K. Pratt	John Wiley, New York
5.	Image processing, analysis and machine	Milan Sonka et al	Vikas Publishing House,

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SYLLABUS

B.Tech. (Information Technology) Sixth Semester

NATURAL LANGUAGE PROCESSING					
Course Code	**106625IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to explore an in-depth understanding of the computational properties of NLP and the commonly used algorithms for processing linguistic information, map the appropriate processing technique to a problem and implement the technique.	**On successful completion of the course, the student will be able to:** **CO1:** Utilize the key concepts of NLP and identify the NLP challenges and issues **CO2:** Develop language modelling for various text corpora across the different languages **CO3:** Illustrate computational methods to understand language phenomena of word sense disambiguation **CO4:** Design and develop applications for text or information extraction/summarization/classification. **CO5:** Apply different Machine translation techniques for translating a source to target language(s).

UNIT I: Introduction to NLP **(CO1)**

What is NLP? Why NLP is Difficult? History of NLP, Advantages of NLP, Disadvantages of NLP, Components of NLP, Applications of NLP, How to build an NLP pipeline? Phases of NLP, NLP APIs, NLP Libraries. **[7 Hrs]**

UNIT II: Language Modelling and Part of Speech Tagging **(CO2)**

Unigram Language Model, Bigram, Trigram, N-gram, Advanced smoothing for language modelling, Empirical Comparison of Smoothing Techniques, Applications of Language Modelling, Natural Language Generation, Parts of Speech Tagging, Morphology, Named Entity Recognition. **[8 Hrs]**

UNIT III: Words and Word Forms **(CO3)**

Bag of words, skip-gram, Continuous Bag-Of-Words, Embedding representations for words Lexical Semantics, Word Sense Disambiguation, Knowledge Based and Supervised Word Sense Disambiguation. **[8 Hrs]**

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UNIT IV: Text Analysis, Summarization and Extraction **(CO4)**

Sentiment Mining, Text Classification, Text Summarization, Information Extraction, Named Entity Recognition, Relation Extraction, Question Answering in Multilingual Setting; NLP in Information Retrieval, Cross-Lingual IR. **[7 Hrs]**

UNIT V: Machine Translation **(CO5)**

Need of MT, Problems of Machine Translation, MT Approaches, Direct Machine Translations, Rule-Based Machine Translation, Knowledge Based MT System, Statistical Machine Translation (SMT), Parameter learning in SMT ((IBM models) using EM), Encoder-decoder architecture, Neural Machine Translation. **[6 Hrs]**

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Chairman (AC)	Chairman (BoS)	Date of Release	Version	

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Speech and Language Processing: An Introduction to Natural Language Processing	Herbert Schildt	Prentice-Hall,
2.	Foundations of Statistical Natural Language Processing	Manning, Christopher D, & Hinrich Schütze	Cambridge

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Natural Language Understanding	James Allen	Pearson Education
2.	Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit	Steven Bird, Ewan Klein, & Edward Loper	O'Reilly Media Publication

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SYLLABUS

B. Tech. (Sixth Semester)

SOLID AND HAZARDOUS WASTE MANAGEMENT					
(Open elective course : Open to all branches)					
Course Code	100641CE	L = 3	T = 0	P = 0	Credits = 3
Examination Scheme	ESE	CT	TA	Total	ESE Duration
	100	20	30	150	3 Hours
	Minimum number of class tests to be conducted = 02		Minimum assignments = 02		

Course Objectives	Course Outcomes
To know source of solid and hazardous wastes. To learn waste generation rates. To understand municipal solid wastes collection system. To understand the labeling and handling of hazardous waste. To learn solid waste disposal.	**On successful completion of the course, the student will be able to:** **CO1:** Describe the solid and hazardous wastes. **CO2:** Explain generation rates of solid and hazardous wastes. **CO3:** Describe handling and segregation of waste at source. **CO4:** Discuss various regulations about the management and handling of hazardous waste. **CO5:** Monitor the solid waste disposal in landfills.

UNIT I: SOURCES OF SOLID AND HAZARDOUS WASTES **CO1**

Sources of solid and hazardous wastes - Need for solid and hazardous waste management - Legislations on management and handling of municipal solid wastes, hazardous wastes, and biomedical wastes. **[6 Hrs.]**

UNIT II: WASTE GENERATION **CO2**

Waste generation rates - Composition - Hazardous Characteristics - TCLP tests - waste sampling- Source reduction of wastes - Recycling and reuse. **[8 Hrs.]**

UNIT III: MUNICIPAL SOLID WASTES COLLECTION **CO3**

Handling and segregation of wastes at source - storage and collection of municipal solid wastes - Need for transfer and transport. **[8 Hrs.]**

UNIT IV: LABELING AND HANDLING OF HAZARDOUS WASTES **CO4**

Waste processing - processing technologies - biological and chemical conversion technologies – Composting, thermal conversion technologies - energy recovery-incineration - solidification and stabilization of hazardous wastes - treatment of biomedical wastes. **[7 Hrs.]**

UNIT V: SOLID WASTES DISPOSAL IN LANDFILLS **CO5**

Site selection - design and operation of sanitary landfills- secure landfills and landfill bioreactors - leachate and landfill gas management. **[7 Hrs.]**

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SYLLABUS

B. Tech. (Sixth Semester)

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Environmental Engineering-II	B.C. Punmia and A.K. Jain	Laxmi Publications,
2.	Introduction to Environmental Engineering and Science	G.M. Masters	Prentice Hall
3.	Manual on Municipal Solid waste management	CPHEEO	Central Public Health and Environmental Engineering Organization, Government of India

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Solid Waste Management	C.L. ell	John Wiley
2.	How to dispose of toxic substances and industrial Waste	P.W. Powers	Noyes Data Corporation
3.	Integrated Solid Waste Management	George Tchobanoglous, Hilary Theisen and Samuel A, Vigil	McGraw- Hill

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SYLLABUS

B.TECH. (SixthSemester)

DATA STRUCTURES AND ALGORITHMS					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	**100642CS**	**L = 3**	**T = 0**	**P = 0**	**Credits =3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 2			**Minimum assignments = 2**	

Course Objective	Course Outcomes
Study and analyze the concepts, features and utilities of various linear/non-linear data structures on the parameters of time and space complexities. Develop the skills of employing the most efficient algorithm and data structure for solving a given computational problem.	**On successful completion of the course, the student will be able to:** **CO1:** Analyze the structure, usage and complexities of operations on different linear data structures. **CO2:** Employ non-linear data structures like Trees and graphs for different application scenarios that represent hierarchical and/or inter-connected data. **CO3:** Apply the algorithms and design techniques to solve problems related to divide and conquer **CO4:** Employ greedy programming and dynamic programming techniques for solving problems **CO5:** Employ backtracking and Branch-and-Bound techniques to solve problems

UNIT –I Linear Data Structures **(CO1)**

Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic notations, Time and Space trade off, Abstract Data Types (ADT). Arrays, Linked lists: Singly Linked List, Doubly List and circular linked list. Standard operations on a Linked List, Stacks and operations, Queue and operations. **[8 hours]**

UNIT – IITrees and Graphs **(CO2)**

Trees, Binary Trees, Binary Search Tree, insertion and deletion, Pre-order, In-order, post-order traversal Heaps, Graphs: Sequential and Linked Representation of Graphs, Adjacency matrices, Adjacency lists, Depth First Search and Breadth First Search, Introduction to Hashing. **[7 hours]**

UNIT – III Divide and Conquer **(CO3)**

Divide and Conquer algorithm design – general idea, Recursion, Recurrence relations, solving recurrence relations using tree method, Binary Search, Quick Sort and Merge Sort as example cases. **[7 hours]**

UNIT – IV Greedy Programming and Dynamic Programming **(CO4)**

Greedy Programming Technique – general idea, Spanning Trees, Minimum Cost Spanning Trees: Prims and Kruskal's algorithm, Dynamic Programing – general idea, Memoization, Warshal's algorithm for All pairs of shortest path, Longest Common Subsequence problem.

. [**7 hours]**

UNIT – V Back Tracking and Branch-and-Bound techniques **(CO5)**
Backtracking – general idea, 4 queen's problem, Graph coloring, Hamiltonian cycle, Branch and Bound – general idea, 8/15 puzzle, 0/1 Knapsack problem. **[7 hours]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Data Structures Using C and C/C++"	Aaron M. Tenenbaum, YedidyahLangsam and Moshe J. Augenstein	PHI
2.	Fundamentals of Data Structures	Horowitz and Sahani	Galgotia Publication
3.	Introduction to Algorithms	Cormen, Lelserson, Rivert	MIT Press / McGraw-Hill
4.	Fundamentals of Computer Algorithm	Ellis Horowitz, SartajSahni, SanguthevarRajasekaran	Galgotia Publications

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	An Introduction to Data Structures with applications	Jean Paul Trembley and Paul G. Sorenson	McGraw Hill
2.	Algorithm Design	Jon Kleinberg and ÉvaTardos	Pearson
3.	Algorithm Design: Foundations, Analysis, and Internet Examples	Michael T Goodrich and Roberto Tamassia	Wiley
4.	NPTEL Link	https://archive.nptel.ac.in/courses/106/102/106102064/	

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SYLLABUS

B.TECH. (Sixth Semester)

UNIX OPERATING SYSTEM					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	100643CS	L = 3	T = 0	P = 0	Credits = 3
Examination Scheme	ESE	CT	TA	Total	ESE Duration
	100	20	30	150	3 Hours
	Minimum number of class tests to be conducted =02			Minimum assignments = 02	

Course Objective	Course Outcomes
Know the architecture and features of unix operating system and be able to use various commands of Unix. Develop shell scripts to perform more complex tasks. To familiarize with the concepts, design, and structure of the UNIX operating system and System calls.	**On successful completion of the course, the student will be able to:** **CO1:**Demonstrate understanding of Unix Architecture, File system and use of basic commands **CO2:** Use advanced commands of Unix OS. **CO3:** Familiarize with basic of Unix shell programming. **CO4:**Demonstrate understanding of basic operating system fundamentals **CO5:** Categorize, compare and make use of Unix System Calls

UNIT –I Introduction to Unix: **CO1**

Architecture of Unix, Features of Unix, Unix file system (Boot Block, Super Block, Inode Block, Data Block), hierarchical structure of Unix File system, Unix Basic Commands –man, echo, passwd, who, date, pwd, cd, mkdir, rmdir, ls, cp, mv, ln rm cat, more, wc, working with text editor, chmode, unmask, ulimit, ps, Instllation of Unix OS. **[8 hours]**

UNIT – II Filters: **CO2**

tail, head, sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, comm, cmp, diff, tr, cpio. grep, egrep, fgrep, sed, awk : Execution, Fields and Records, Scripts, Operations, Patterns, Actions, Associative Arrays, String Functions, Mathematical Functions. **[7 hours]**

UNIT – III Shell Programming: **CO3**

Ordinary and environment variables. read commands. Command line arguments. Set and Unset a variable. Using expr, control statements: if, while, until, for and case control statements. Simple shell program examples. **[7 hours]**

UNIT – IV: The buffer cache: **CO4**

buffer header, structure of the buffer pool, **Internal representation of files**: inodes, Structure of Regular File, Directories, Conversions of a Path name to an INODE, Super Block, INODE assignment to a New File. **[7 hours]**

UNIT – V System Calls: **CO5**

File Manipulation - open(), read(), write(), close(), Process states and transitions, Process Control- fork, vfork, exit, wait, Communication - chmod(), umask(), chown() **[7 hours]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Advanced UNIX: A Programming's Guide	S. Prata	BPB Publications, New Delhi.
2.	Design of UNIX Operating System	M.J. Bach	Prentice Hall of India.
3.	Your Unix the ultimate guide	Sumitabha Das	TMH. Edition.

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Unix shell programming	Y.Kanetkar	BPB Pub
2.	Unix and shell Programming	Behrouz A. Forouzan, Richard F. Gilberg Thomson	Thomson Asia
3.	NPTEL Link	https://archive.nptel.ac.in/courses/117/106/117106113/	

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SYLLABUS

B.Tech. (Sixth Semester)

Renewable Energy Engineering					
(Open Elective Course : Open to all branches except EE)					
Course Code	**100644TE**	**L =3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To describe in brief renewable energy sources and their usage. To create awareness among budding engineers regarding common renewable energy sources as finiteness of fossil fuel reserves and large scale environmental degradation caused by their widespread use, particularly global warming, urban air pollution and acid rain, strongly suggests that harnessing of non-conventional, renewable and environment friendly energy resources is vital for steering the global energy supplies towards a sustainable path.	**On successful completion of the course, the student will be able to:** **CO1:** Discuss different sources of renewable energy and it's fundamentals. **CO2:** Explain solar energy and its utilization using different collectors and implementation in solar plant. **CO3:** Describe the concepts involved in wind energy systems, its components, types and performance. **CO4:** Discuss concepts involved in Bio-mass energy and conversion technology using different plants. **CO5:** Explain principles of thermo electric power and Fuel cells using hydrogen energy.

UNIT – I **CO1**

Introduction: Overview of various renewable energy sources, limitations of non-renewable energy sources, Need, availability, classification, advantages & disadvantages of different renewable sources. Energy storage, distribution and conservation. **[7Hrs]**

UNIT – II **CO2**

Solar Energy Fundamentals: Solar Cells; Theory of Solar Cells, Materials, Solar Thermal Energy: Solar collectors and its types, Applications of solar collectors, storage, Solar Thermal Power Plants, merits/demerits. **[8Hrs]**

UNIT – III **CO3**

Wind Energy Fundamentals: Basic Principles of Wind Energy conversion, Site Selection criterion, Types of Rotors, Characteristics, performance & limitations of energy conversion systems. **[8Hrs]**

UNIT – IV **CO4**

Bio-Mass Energy Fundamentals– Conversion Technology, Classification of Plants, Advantages & Disadvantages. **[6 Hrs]**

UNIT –V **CO5**

Thermo Electric Power – Basic Principles, Thermo Electric Materials, Performance & Limitations**.**

Hydrogen Energy – Principles of conversion, Methods of H_2 production.

Fuel cells: principle, types. **[7Hrs]**

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Text Books:

S. No.	Title	Author(s)	Publisher
1.	Non-Conventional Energy Sources	G.D. Rai	Khanna Publishers.
2.	Non-Conventional Energy Sources	B.H. Khan	McGraw Hill Education India Private Limited

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Solar Energy	S.P. Sukhatme	McGraw Hill Education
2.	Energy Conversion Systems	Rakosh Das Begamudre	New Age International Publishers

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SYLLABUS

B.Tech. Sixth Semester

ELECTRIC VEHICLES					
(Open Elective Course : Open to all branches)					
Course Code	**100645EE**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To familiarize the students about the fundamentals of electric vehicles (EV), their architecture, technologies and to discuss energy storage with charging techniques.	**On successful completion of the course, the student will be able to:** **CO1:** Explain the infrastructure of Electric Vehicles. **CO2:** Explain the Drive Cycle and various parameters of battery. **CO3:** Estimate the SoH and SoC of EV battery. **CO4:** Explain the Thermal, Mechanical as well as Electric design of battery pack. **CO5:** Evaluate the performance of Electric vehicle.

UNIT-I **CO1**

Introduction to Electric Vehicles: Overview of Electric Vehicles in India, E V Program in India, Batteries, Charging and Swapping Infrastructure, Procurement of Lithium for batteries, EV Subsystems, Forces acting on moving a vehicle, Aerodynamic drag, Rolling Resistance and Uphill Resistance, Power and Torque for acceleration. **[6Hrs]**

UNIT – II **CO2**

Drive Cycle and EV Subsystem: Drive Cycle and Energy used per km, EV Subsystem: Design of EV Drive Train, Introduction to Battery Parameters, Importance of Lithium Ion Battery, Batteries in Future, Li-Ion Battery Cells. **[9Hrs]**

UNIT - III **CO3**

Battery Management System: State of Health(SoH) and State of Charge(SoC) estimation and Self Discharge, Battery Pack Development, Computation of Effective cost of battery, Charging Batteries.. . **[6Hrs]**

UNIT – IV **CO4**

Battery Pack Design: Fundamentals of Battery Pack Design, Mechanical Design, Thermal Design, Electrical Design, BMS Design of Electric Vehicle, EV Motors and Controllers - Understanding Flow. . **[7Hrs]**

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UNIT – V **CO5**

Performance Parameters and Chargers: Power and Efficiency, Torque Production, Speed and Back EMF, Future Frontiers, EV Chargers- Slow and Fast, Public Chargers, Economics of Public Chargers. **[7Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Electric & Hybrid Vehicles - Design Fundamentals	Iqbal Hussain	CRC Press
2.	Electric Vehicle Technology Explained	James Larminie	John Wiley & Sons.

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Modern electric, hybrid electric, and fuel cell vehicles: fundamentals, theory & design	Ehsani, Mehrdad, Yimin Gao, and Ali Emadi.	CRC Press

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SYLLABUS

B.Tech. (Sixth Semester)

IoT Applications					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	**100646ET**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
This course aims to provide knowledge of fundamentals of IoT with its architecture, protocols and applications. It covers the development of Internet of Things (IoT) for sensing, actuating, processing & communication.	**On successful completion of the course, the student will be able to:** **CO1:** Demonstrate an understanding of the various concepts, terminologies and architecture of IoT systems. **CO2:** Use sensors and actuators for interfacing with IoT. **CO3:** Demonstrate an understanding of various applications of IoT. **CO4:** Apply various protocols & communication APIs for design of IoT systems to connect IoT related technologies. **CO5:** Analyze different network topologies & networking with APIs for IoT.

UNIT-I **CO1**

INTRODUCTION TO IOT: Introduction, Definition and Characteristics of IoT, Things in IoT, IoT Stack, Enabling Technologies, IoT Challenges, IoT Functional Blocks. **[6Hrs]**

UNIT – II **CO2**

IOT SENSORS NETWORKS: Sensor Interfacing, Working of IoT Sensors, Types of IoT Sensors & Actuators, Microcontroller in IoT, Role of Embedded system in IoT. **[7Hrs]**

UNIT - III **CO3**

DOMAIN SPECIFIC IOT: Illustrated domains: Smart Cities, Environment, Energy, Agriculture, Health and Lifestyle issues, Home Automation, Industry Application. **[7Hrs]**

UNIT – IV **CO4**

DESIGN OF IOT: IoT Protocols, IoT Communication Models, IoT Communication APIs: REST &WebSocket based (Introduction and Block diagram). **[5Hrs]**

UNIT – V **CO5**

IOT AND M2M: Block diagram of M2M, IoT Vs M2M, Simplified OSI model, Network Topologies, Types of Internet Networking and Network APIs. **[5Hrs]**

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Text Books:

S. No.	Title	Author(s)	Publisher
1.	Internet of Things A Hands-On-Approach	Vijay Madisetti, ArshdeepBahga	Orient Blackswan Private Limited.

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Foundational Elements of an IOT Solution - The Edge, Cloud and Application Development	Joe Biron & Jonathan Follett	Oreilly media inc.
2.	The Internet of Things (A Look at Real World Use Cases and Concerns)	Lucas Darnell	International Kindle paper white Edition.
3.	The Internet & Its Protocol-A comparative Approach	Adrian Farrel	Morgan Kaufmann Publishers.
4.	Internet of Things-Architecture and Design Principles	Raj Kamal	McGraw Hill Education (India) Private Limited.
5.	Introduction to IoT	S. Misra, A. Mukherjee, and A. Roy	Cambridge University Press.
6.	Introduction to Industrial Internet of Things and Industry 4.0.	S. Misra, C. Roy, and A. Mukherjee	CRC Press.

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SYLLABUS

B.Tech.(Sixth Semester)

Programming in PHP and MySQL					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	**100647IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The objective of this course is to make students aware about the concepts and basics of web development using PHP. This syllabus also aims to understand the database using my SQL and helps students to develop user form.	**On successful completion of the course, the student will be able to:** **CO1:** Demonstrate an understanding of the concepts of PHP, variables and operators. **CO2:** Demonstrate an understanding of arrays and functions used in PHP **CO3:** Work with files and directories **CO4:** Create and manage sessions and cookie. **CO5:** Implement web pages in order to store and retrieve data using database like MySQL

UNIT – I **CO1**

Introduction to PHP: Evaluation of Php, XAMP/LAMP installation, Basic Syntax, echo function, Defining variable and constant, PHP Data type, Operator and Expression. Making Decisions, Doing Repetitive task with looping, Mixing Decisions and looping with Html. **[7hrs]**

UNIT II **CO2**

Arrays and Functions: Anatomy of an Array, Creating index based and Associative array Accessing array, Element Looping with Index based array, Looping with associative array using each() and foreach(), Some useful Library function. PHP string functions. **[7hrs]**

UNIT – III **CO3**

Working with files and Directories: Working with file and Directories Understanding file& directory, Opening and closing, a file, Coping, renaming and deleting a file, working with directories, Creating and deleting folder, File Uploading & Downloading. GET and POST variables. Embedding PHP. Sending Email. **[8hrs]**

UNIT – IV **CO4**

Session and Cookie: Introduction to Session Control, Session Functionality What is a Cookie, Setting Cookies with PHP. Using Cookies with Sessions, Deleting Cookies, Registering Session variables, Destroying the variables and Session. MD5 encryption, Exception Handling. **[8hrs]**

UNIT – V **CO5**

Exception Handling, Database Connectivity with MySQL:

Understanding Exception and error, Try, catch, throw, Error tracking and debugging.

Introduction to RDBMS, Connection with MySql Database, Performing basic database operation(DML) (Insert, Delete, Update, Select), Setting query parameter, Executing query-Join (Cross joins, Inner joins, Outer Joins, Self joins.)

User Regsitration, Login, password change. **[10hrs]**

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Text Books:

S. No.	Title	Author(s)	Publisher
1.	Learning PHP, MySQL & JavaScript with j Query, CSS & HTML5	Robin Nixon	O'Reilly, Shroff Publishers & Distributers Private Limited (SPD)
2.	PHP, MySQL & JavaScript All-in-One For Dummies	Richard Blum	Wiley

Video Reference:
PHP and MySQL, Spoken-Tutorial IIT Bombay,
URL: https://www.youtube.com/playlist?list=PL7WFbgpeASD3kmW7h5qhGZGIWTOXGGbTZ

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SYLLABUS

B.Tech. (Sixth Semester)

OPERATION RESEARCH					
(Open Elective Course : Open to all branches)					
Course Code	**100648ME**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The objective of this course is to impart an understanding to mathematical tools that are needed to solve optimization problems.	**On successful completion of the course, the student will be able to:** **CO1:** Formulate and solve real-world problems as linear programs for better decision-making. **CO2:** Solve specialized linear programming models like the transportation and assignment Models. **CO3:** Model a dynamic system as a queuing model and compute important performance measures. **CO4:** Use CPM and PERT techniques, to plan, schedule and control project activities. **CO5:** Propose the best strategy using decision making methods under game theory & applyconcepts of Simulation to optimize practical problems.

UNIT – I **(CO1)**

Introduction: Various stages of O.R., Fields of application, optimization and its classification. **General Linear Programming Problems**- Introduction, maximization and minimization of function with or without constraints, formulation of a linear programming problem, graphical method and simplex method, Big M method, degeneracy, application of L.P.P. in Mechanical Engineering.**[8Hrs]**

UNIT – II **(CO2)**

The Transportation Problems: Mathematical formulation computational procedures, Modified Distribution Method, Vogel's Approximation Method, Solution of balanced and unbalanced transportation problems and case of Degeneracy. **[6Hrs]**

The Assignment Problems: Mathematical formulation of assignment problems, solution of assignment problems, traveling salesman problems, Aircrew Assignment problems. **[6Hrs]**

UNIT – III **(CO3)**

Waiting Line Theory: Basic queuing process, basic structure of queuing models, some commonly known queuing situations, Kendall's notation, solution to M/M/1:∞ /FCFS model. **[5Hrs]**

UNIT – IV **(CO4)**

Network Analysis: CPM & PERT: Network Representation, Techniques for drawing network. Project cost, Optimum project duration, project crashing and Time estimation in PERT. **[5Hrs]**

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UNIT-V **(CO5)**

Game Theory: Introduction, two-person zero sum game, methods for solving two-person zero sum game: when saddle point exists, when no saddle point exists, solution of 2xn and mx2 game-Method of subgame, Graphical method. **[3Hrs]**

Simulation: Basic concept of simulation, application of simulation, Merits and Demerits of simulation, Monte-Carlo simulation, simulation of Queuing system**.** **[3hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Operation Research	Hira & Gupta	S.Chand & Co.
2.	Operation Research	N.D.Vohra	TMH, New Delhi

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Operation Research	S.D.Sharma	S.Chand & Co.
2.	Operation Research	Hamdy.M. Taha	TMH, New Delhi
3.	Operation Research	J K Sharma	Macmillan
4.	Operation Research	A. P. Verma	S. K. Kataria & Sons

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SYLLABUS

B.Tech. (Sixth Semester)

FINANCIAL MANAGEMENT					
(Open Elective Course : Open to all branches)					
Course Code	**100649mg**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
1. The objective of this course is to understand various concepts related to financial management. 2. To study in detail various tools and techniques in the area of finance. 3. To develop the analytical skills that would facilitate financial decision making.	**On successful completion of the course, the student will be able to:** **CO1:** Classify funding sources and demonstrate knowledge of value of moneyovertime. **CO2:** Examine complexities associated with financing decision. **CO3:** Demonstrate understanding of the concept of budgeting and evaluate proposals. **CO4**: Select and apply techniques in management of working capital. **CO5:** Interpret the profit distribution decisions.

UNIT – I Introduction to Financial Management **CO1**

Financial Management: Nature and Objectives, Profit maximization v/s Wealth maximization, Finance Function, Time value of money - Discounting and Compounding Techniques, Long term and Short-term sources of Finance, Introductionto Capital Market. **[8 HRS]**

UNIT – II Financing Decision **CO2**

Cost of Capital: Weighted Average Cost of Capital

Capital Structure: Factors, Approaches and Theories

Leverage: Operating and Financial Leverage: Impact, Trading on Equity **[7 HRS]**

UNIT – III Investment Decision **CO3**

Budget: Concept and Types, Budgetary Control, Capital Budgeting, Zero based Budgeting.

. **[7 HRS]**

UNIT – IV Working Capital Decision **CO4**

Management of Working Capital: Concept, Need, Factors and Estimation of WorkingCapital, Inventory and Receivables Management, Management of Cash **[7 HRS]**

UNIT – V Dividend Decision **CO5**

Dividend Policy: Types, Factors Influencing Dividend Policy and Dividend Models. **[7 HRS]**

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Text Books:

S. No.	Title	Author(s)	Publisher
1.	Financial Management	Eugene F Brigham	CengageLearning
2.	Financial Management	M Y Khan, P K Jain	Mc Graw Hill

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Financial Decision Making: Concepts, Problems and Cases	John J. Hampton	PHI
2.	Financial Management and Policy	V. K. Bhalla	Anmol Publications
3.	Financial Management	Tulsian & Tulsian	S Chand
4.	Essentials of Financial Management	I M Pandey	Vikas Publishing House
5.	Financial Management	P. Chandra	Mc Graw Hill

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SYLLABUS

B.Tech. (Sixth Semester)

PYTHON					
(Open Elective Course : Open to all branches)					
Course Code	**100650CA**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
1. To understand basic concepts of python, Conditions & branching, String operations. 2.To familiarize student with syntax and implementation of Tuples and Dictionary. 3.To make students understand Objects &Classes. 4. To provide detailed knowledge Data Handling with Pandas.	**On successful completion of the course, the student will be able to:** **CO1:** Demonstrate an understanding of concepts of python, String operations. **CO2:** Implement program related to tuples and dictionary. **CO3:** Design program related to functions. **CO4:** Design good types of graphs in Python. **CO5:** Demonstrate an understanding of the basic concept of panda and NumPy.

UNIT-I **CO1**

Python basic, Data types, Typecasting, Data type conversions, String operations, Slicing, Stride, String Methods. Python programming fundamentals-conditions & Branching. [**8Hrs]**

UNIT-II **CO2**

Tuple, Lists, Set, Dictionaries- Tuple indexing, Slicing, Nesting, List indexing, List functions-extend, append, delete, split. List Aliasing, List Clone. Set –creating set, Set Operations. Dictionaries. **[7Hrs]**

UNIT-III **CO3**

Python Functions: Defining a function, Calling a function, Types of functions, Function Arguments, Anonymous functions, Global and local variables, Organizing python codes using functions. **[7Hrs]**

UNIT-IV **CO4**

Introduction to matplotlib. Plotting different types of graphs using Python. Image processing in Python. Reading images in python. Some basic operations with images: Resizing, tilt, rotation. **[7Hrs]**

UNIT-V **CO5**

Data Handling with PandasCreation of a panda's series from a dictionary of values and array Creation of a panda's series from a dictionary of values and array, Importing and exporting data between pandas and CSV file, Basic concepts and use of NumPy. **[7Hrs]**

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TextBooks:

S. No.	Title	Author(s)	Publisher
1.	Python Programming	Anurag Gupta,G P Biswas	Mc Graw Hill
2.	Complete Reference Python	Martin C. Brown	Mc Graw Hill
3.	Python Programming for absolute beginners	Michal Dawson	Cengage Learning

ReferenceBooks:

S. No.	Title	Author(s)	Publisher
1.	Python for beginners	Harsh Bhasin	New Age International Publisher
2.	Python for beginners	Prof. Rahuk E. Borate Dr. Sunil Khilari Prof. Rahul S Navale	Shroff/X-Team

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SYLLABUS

B.Tech. (Information Technology) SixthSemester

PRINCIPLES OF COMMUNICATION SYSTEM LAB					
Course Code	**106691IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
The objective of this course is to impart understanding of various analog modulation and demodulation, analog to digital conversion and various digital and advance modulation techniques.	**On successful completion of the course, the student will be able to:** **CO1**: Identify different essential sub-systems of communication system. **CO2:**Demonstrate the concepts of analog transmission/ reception using modulators/demodulators. **CO3:** Visualize different processes of analog to digital conversion by performing the experiments for sampling Quantization and Coding. **CO4:** Apply the concepts of digital transmission/ reception using modulators/demodulators. **CO5:** Compare BPSK, QPSK & QAM signal constellation, ASK, FSK BPSK & DPSK digital modulation schemes.

Suggested List of Experiments (but should not be limited to):

1. To Draw the O/P waveform of Amplitude Modulation & Demodulation & Calculate Modulation Index.
2. To Draw the O/P waveform of Frequency Modulation & Demodulation & Calculate Modulation Index.
3. To Study DSB Transmitter & Receiver.
4. To Study SSB Transmitter & Receiver.
5. To Study FM Transmitter & Receiver.
6. To Observe & plot the Graph of PAM Modulation & Demodulation.
7. To Observe & plot the Graph of PPM Modulation & Demodulation.
8. To Observe & plot the Graph of PWM Modulation & Demodulation.
9. To Perform Sampling & Reconstruction of original signal & to calculate the Sampling Frequency.
10. To Perform Amplitude Shift Keying (ASK) thereby determining relative change in Amplitude.
11. To Perform Frequency Shift Keying (FSK) thereby determining relative change in Frequency.
12. To Perform Phase Shift Keying (PSK) thereby determining relative change in Phase
13. To Perform Quardrature Phase Shift Keying (QPSK) thereby determining relative change in Phase.
14. To Perform Quardrature Amplitude Modulation (QAM).
15. To perform Adaptive Delta Modulation, Demodulation.
16. To perform Delta Modulation & Compare it with Adaptive Delta Modulation (ADM).
17. To study & perform Transmission & Reception of signal using TDM Technique.

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Principles of Communication system	H.Taub and D.L. Shiling	TMH
2.	Communication Systems https://elib4u.ipublishcentral. com/pdfreader/communication-systems	S L Kakani and Priyanka Punglia	New Age International Publisher
3.	Principles of Electronic Communication Systems	Louis E. Frenzel Jr.	McGraw Hill Higher Education

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Electronic Communications	Roddy&Coolen	PHI
2.	An Introduction to the Principle of Communication Theory	J.C. Hancock	McGraw Hill
3.	Communication System	A.B. Carlson	McGraw Hill
4.	Electronic Communication System	Kenedy& Davis	TMH

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SYLLABUS

B.Tech. (Information Technology) Sixth Semester

ARTIFICIAL INTELLIGENCE LAB					
Course Code	**106692IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
The course objective is to make the student understand the concept of AI problems and show proficiency in traditional AI languages including ability to write simple to intermediate programs, ability to understand code written in that language and to discuss various techniques and algorithms of AI used in general problem solving, optimization problems, constraint satisfaction problems, and game programming.	**On successful completion of the course, the student will be able to:** **CO1**: Analyze AI problem and write PROLOG / LISP syntax to solve it. **CO2:** Draft production rules (iterative / recursive) for an AI problem. **CO3**:Demonstrate how to use different data structures for solving AI problems **CO4**: Decide upon situations that crop up such as syntax / compile-time / run-time errors. **CO5:** Simulate AI problems by using generalized solutions in PROLOG / LISP syntax.

LIST OF EXPERIMENTS

1. Write a prolog program to find the rules for parent, child, male, female, son, daughter, brother, sister, uncle, aunt, ancestor given the facts about father and wife only.

2. Write a program to find the length of a given list

3. Write a program to find the last element of a given list

4. Write a program to delete the first occurrence and also all occurrences of a particular element in a given list.

5. Write a program to find union and intersection of two given sets represented as lists.

6. Write a program to read a list at a time and write a list at a time using the well defined read & write functions.

7. Write a program given the knowledge base,

If x is on the top of y, y supports x.

If x is above y and they are touching each other, x is on top of y.

A cup is above a book. The cup is touching that book. Convert the following into wff's, clausal form; Is it possible

To deduce that `The book supports the cup'.

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8. Write a program given the knowledge base,

If Town x is connected to Town y by highway z and bikes are allowed on z, you can get to y from x by bike.

If Town x is connected to y by z then y is also connected to x by z.

If you can get to town q from p and also to town r from town q, you can get to town r from town p.

Town A is connected to Town B by Road 1. Town B is connected to Town C by Road 2.

Town A is connected to Town C by Road 3. Town D is connected to Town E by Road 4.

Town D is connected to Town B by Road 5. Bikes are allowed on roads 3, 4, 5.

Bikes are only either allowed on Road 1 or on Road 2 every day. Convert the following into wff's, clausal form and deduce that `One can get to town B from town D'.

9. Solve the classical Water Jug problem of AI.

10. Solve the classical Travelling Salesman Problem of AI.

Laboratory Equipments / Machine Requirements: Windows / Linux operating system, 4GB RAM, 5GB free disk space and Python 3.6 or higher version

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Logic & prolog programming	Ivan Bratko	Addison-Wesley
2.	Introduction to Turbo Prolog	Carl Townsend	BPB, Publication
3.	Programming in PRLOG	W.F. Clocksin&Mellish	Narosa Publication House

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SYLLABUS

B.Tech. (Information Technology) Sixth Semester

ANDROID LAB					
Course Code	**106693IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
The objective is to understand the working of Android applications and to create GUI and handle events in Android applications with data storage, APIs and Databases.	**On successful completion of the course, the student will be able to:** **CO1:** Apply techniques to develop applications for the Android phone. **CO2**: Use SQLite for database connectivity with the Android Application. **CO3:** Demonstrate API connections to App. **CO4:** Develop Android applications using Android Studio. **CO5:** Publish Android applications to the Android Market.

1. Download and setup Android Environment

2. Using the Development environment

a. Create a new Project using wizard

b. Add source and resource files.

c. Import existing projects into workspace

d. Create testing Emulator

e. Compile and run the project

f. Debug the project

g. Debug on android device.

3. XML Files

a. AndroidManifest.xml

a.i. Edit the manifest and change min sdk and target sdk of application.

a.ii. Add main activity entries in manifest.

a.iii. Add second activity entries in manifest.

a.iv. Add Entries for Service, Broadcast receivers.

a.v. Add uses permissions for reading files, internet, camera.

b. Layouts

b.i. Create Linear Layout in xml

b.ii. Create Relative Layout in xml

b.iii. Create frame layout in xml

b.iv. Create a complex mixed layout using all above layouts

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c. Drawables
c.i. Create xml drawable for rectangular, oval and other basic shapes
c.ii. Create xml drawable with Layer list for complex shapes.
d. Values
d.i. Create strings.xml to store all your application strings.
d.ii. Create color.xml to store all your color values
d.iii. Create styles.xml to store all your custom themes and style objects
e. Alternate resources based on qualifiers
e.i. Create separate drawables folders and xml files based on screen density (LDPI, MDPI, HDPI, XHDPI, XXHDPI)
e.ii. Create separate styles.xml based on different android versions.
e.iii. Create separate layout folders based on device screen sizes and orientations.

4. Creating User Interface
a. Create application with Basic Views (Text view, Button, List View)
b. Create application with different Layouts (Linear, Relative, Frame)
c. Create application to handle and respond on click using Click Listeners

5. Assets and Images
a. Create application which will access files from Assets folder (Images, sounds, Custom Fonts)

6. Application Fundamentals
a. Activities
a.i. Create application with one activity and display a layout created in xml.
a.ii. Create application which will log all activity lifecycle events using Android log api.
a.iii. Create application which should be Saving and restoring app state (eg text view text, checkbox checked state)
b. Intents
b.i. Create application which will start another activity using intent.
b.ii. Create an activity which will pass data to second activity using intent.
b.iii. Create activity which will start second activity and get response back from second activity.
c. Services
c.i. Create

7. Content Providers
a. System provided content providers
a.i. Create application which can access/modify Contacts of device.
a.ii. Create application which can access & display Images available on device.
a.iii. Create application which can access and play Media files (Audio & Video)
b. Custom Contact providers
b.i. Create application which will provide some data to other applications using Content Provider system.

8. Broadcast Receivers
a. Create application to Listen to following system events using Receivers
a.i. Incoming SMS
a.ii. In and outgoing Phone Call
a.iii. Low Battery
a.iv. Storage state changed
b. Create application which will broadcast Custom event to custom Receivers.

9. Create application which will display following Notifications
a.i. Toast notification
a.ii. Status bar notification
a.iii. Dialog notification

10. Preference & Data Storage
a. Create application which will save and read back data using Shared Preference
b. SQLite database
b.i. Create app to create database using Open helper
b.ii. Create app to read, write and delete database entries

11. Networking & Web API
a. HTTP connectivity
a.i. Create app to connect and fetch data from a Http server/ website using URLConnection
a.ii. Create app to connect and fetch data from a Http server/ website using HTTPClient library
a.iii. Create app to connect and post data to Http server/ website using URLConnection
a.iv. Create app to connect and post data to Http server/ website using HTTPClient library
b. TCP Sockets or Sockets
b.i. Create a server app using tcp socket, it will send “Welcome” to client when its connected.
b.ii. Create a client app using tcp socket, it will send “Hello” to server once connected.

12. Google API

a. Create application using Maps api, it should display marker on current location of user

b. Create application which will display ads using Admobapi

13. Accessing android hardware

a. Create Application to take picture and save it to file storage using camera api

b. Create application to display current direction using sensor api

c. Create application to show a toast if phone is waved in air.

d. Create application to show list of paired and nearby bluetooth devices.

14. Facebook SDK

a. Create application which can share link on facebook using Facebook sdk.

b. Create application which can share photo on facebook using Facebook sdk.

15. Publish to playstore

a. Enable Obfuscation for your application using Proguard

b. Export Signed application package

c. Prepare Store listing

d. Upload and publish apk

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Head First Android	Jonathan Simon	O'Reilly Media, Inc
2.	Android Application Development	Reto Meier	Wiley Publishing , Inc

Reference Books:-

S.No.	Title	Author(s)	Publisher
1.	Beginning Android 4 Application Development	Wei-Meng Lee	John Wiley & sons Inc

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SYLLABUS

B.Tech. (Information Technology) Sixth Semester

MINOR PROJECT – II					
Course Code	**106694IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	50	**-**	**25**	**75**	**-**

Course Objectives	Course Outcomes
• To provide an opportunity for students to understand real-life practices and implementation of theoretical Engineering knowledge in practical environment. • To provide an opportunity for students to acquire interpersonal skills and ability for team work. • To obtain knowledge of how to make optimal decisions to resolve technical challenges.	**On successful completion of the course, the student will be able to:** **CO1:** Apply engineering knowledge and utilize technical resources in real life projects by integrating classroom theory with real-time situations. **CO2:** Acquire practical skills, organizational skills, Communication skills, professional awareness and experience working on projects. **CO3:** Predict by making optimal decisions to resolve technical challenges. **CO4:** Write technical documents related to the project work completed. **CO5:** Demonstrate oral presentations of the project in an efficient manner.

- Minor project is introduced in the sixth semester with a specific objective to strengthen the understanding of student's fundamentals through effective application of theoretical concepts of Software Engineering and Project Management.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- Student Groups with 3/4 members should identify a topic of interest in consultation with Faculty Guide.
- Review the literature and gather information pertaining to the chosen topic.
- State the objectives and develop a methodology to achieve the objectives. Develop an SRS.
- Carry out the design/ develop codes/programs to achieve the objectives.
- Demonstrate the novelty of the project through the results and outputs.
- The progress of the project is evaluated based on a minimum of two reviews.
- The review committee may be constituted by the Head of the Department.
- A project report is required at the end of the semester.
- The product has to be demonstrated for its full design specifications.
- The internal evaluation will be made based on the product, the report and a viva-voce examination, conducted internally by a committee appointed by the Head of the Department comprising HoD or a Project coordinator for that program.

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SYLLABUS

B.Tech. (Common to All Branches) Sixth Semester

ESSENCE OF INDIAN KNOWLEDGE TRADITION					
Course Code	**100695HM**	**L = 0**	**T = 0**	**P = 0**	**Credits = -**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	-	**-**	**25**	**25**	**-**

Course Objectives	Course Outcomes
The course aims to impart the ability to understand, connect up and explain basics of Indian traditional knowledge in modern scientific perspective.	**On successful completion of the course, the student will be able to:** **CO1:** Demonstrate awareness about philosophy of Indian culture and scriptures **CO2:** Exhibit understanding about Indian perspective of modern scientific world-view and basic principles of Yoga and holistic health care. **CO3:** Display understanding about the religion and major philosophical traditions in India. **CO4:** Exhibit acquisition of fine arts and technology in India. **CO5:** Demonstrate understanding about traditional educational systems in India.

UNIT – I **(CO1)**

Introduction to Indian Ethos: Indian Culture; Civilization; Heritage; General characteristics and major literature; Vedic visions and Upanishads

UNIT – II **(CO2)**

Modern Science and Indian Knowledge System: Development of Science and Technology in ancient, medieval and modern India; Eminent Scientists; Yoga and Holistic health care

UNIT – III **(CO3)**

Religion and Philosophy: Religions practiced in India and understanding their philosophy; Philosophical tradition in India.

UNIT – IV **(CO4)**

Art & Technology: Fine arts & Linguistics; Economics & Mathematics; Architecture & Agriculture; Astronomy & Astrology; Ayurveda, Metallurgy & Chemistry

UNIT – V **(CO5)**

Education System in India: Goals of traditional knowledge system; Gurukuls & Student-Teacher relation; Renowned universities, Traditional knowledge & Intellectual Property

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Text Books:

S.No.	Title	Author(s)	Publisher
1.	Text and Interpretation: The India Tradition	Kapil Kapoor	D K Printworld
2.	Modern Physics and Vedant	Swami Jitatmanand	Bharatiya Vidya Bhavan
3.	Essentials of Indian Philosophy	M Hiriyanna	Motilal Banarsidass

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Science of Consciousness Psychotherapy and Yoga Practices	RN Jha	Vidyanidhi Prakasham
2.	Yoga-darshanam with Vyasa Bhashya	GN Jha	Vidyanidhi Prakasham
3.	Founders of Sciences in Ancient India	Satya Prakash	Vijay Kumar Publisher